Architecture and Methodology

Problem 6 — Rule-Based POS Tagger for Technical Repair Audio Transcripts

1. Problem statement

Technical repair transcripts contain domain-specific words that can have different grammatical roles depending on context. For example, “pump” is a noun in “the pump is broken” and a verb in “pump the water line.” The goal is to assign a part-of-speech (POS) label to each token using transparent, hand-written rules.

2. System architecture

Raw transcript → sentence segmentation/tokenization → ordered RegexpTagger cascade → DefaultTagger fallback → contextual correction rules → tagged tokens → comparison with gold annotations.

3. Main components

Preprocessing: `preprocessing.py` splits text into sentences and tokens. It also includes optional simple lemmatization and stopword removal helpers. Stopword removal is not applied before POS tagging because articles and prepositions are useful context.

Regex cascade: `models.py` uses NLTK’s `RegexpTagger`. Patterns cover punctuation, numbers, determiners, pronouns, auxiliary verbs, prepositions, conjunctions, known adjectives, known verbs, and suffix clues such as -ly, -ing, and -ed.

Fallback: `DefaultTagger('NOUN')` labels a token when no regular expression matches. This is a simple fallback, not a guarantee that the token is a noun.

Context correction: a second pass checks neighboring words. A determiner before an ambiguous technical word favors a noun; a modal or sentence-initial command pattern can favor a verb.

4. POS tag inventory

Tag	Meaning	Example
NOUN	Object/person/thing	pump, wire, motor
VERB	Action	replace, inspect
ADJ	Describes a noun	damaged, loose
ADV	Describes manner	slowly, carefully
DET	Determiner/article	the, a
PRON	Pronoun	they, it
ADP	Preposition	near, with
AUX	Auxiliary/modal	is, will
CONJ	Conjunction	and, but
NUM	Number	12
PUNCT	Punctuation	. , ?

5. Rule ordering

Patterns are evaluated in order, so specific patterns should precede broad patterns. For example, a known adjective such as “damaged” should be tagged before the general -ed suffix rule. If a catch-all pattern is placed first, later rules may never run.

6. Evaluation method

The project compares predicted tags with manually supplied gold tags in `data/annotated/gold_standard.jsonl`. Token accuracy is:

```
Token accuracy = (number of correctly tagged tokens) / (total number of gold-standard tokens)
```

This is a rule-based system and has no model-training stage. The evaluation is a small demonstration benchmark; a larger, independently annotated test set is needed for a realistic estimate of performance.

7. Limitations

• Ambiguous words can still be misclassified in less common sentence structures.
• Suffixes are clues, not definitive grammatical evidence.
• The default NOUN fallback can mislabel unseen verbs or adjectives.
• Domain vocabulary and language variation require ongoing rule updates.
• A small hand-built test set may overstate generalization.